

Empirical Validation of Dynamic Graph-Agent Economic Models: Out-of-Sample Shock Propagation in Global Production Networks

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Abstract

A central promise of multi-agent and graph-theoretic economic frameworks is that explicitly modeling network topology, heterogeneous agent policies, and partial observability yields superior predictions of systemic shocks and non-equilibrium dynamics compared to aggregate macroeconometric models. However, theoretical formalisms frequently lack rigorous empirical falsification protocols. In this paper, we operationalize a dynamic graph-agent architecture into an out-of-sample prediction benchmark on the global economy using the OECD Inter-Country Input-Output (ICIO) database (1995–2022), covering 80 economies and 50 industries ($\approx 4,000$ country-industry nodes). We test whether a dynamic relational graph model can predict cross-sectional output growth and tail-risk contractions during the held-out 2020 COVID-19 production shock, conditional on pre-shock network topology (G_{2019}) and common aggregate shock signals (z_{2020}). Evaluating against six conventional macroeconometric and input-output baselines—including Panel Autoregressions, Dynamic Factor Models, Leontief Input-Output Multipliers, Centrality Regressions, and Spatial Autoregressions—we show that the proposed Graph/Agent architecture achieves lower out-of-sample Root Mean Squared Error (RMSE = 0.0412 vs. 0.0587 for Panel AR and 0.0521 for Leontief) and superior tail-risk discrimination (AUPRC = 0.684 vs. 0.412 for Leontief). An extensive 8-component ablation study demonstrates that asymmetric directed edge flows, non-linear message passing, and heterogeneous agent state vectors are the primary drivers of predictive gains. Finally, we establish a formal prediction-observation divergence criterion $\Delta(t)$ to govern model falsification and iterative structural refinement.

Keywords: Production Networks, Graph Neural Networks, Multi-Agent Systems, Out-of-Sample Forecasting, Systemic Risk, OECD ICIO, Input-Output Analysis.

JEL Classification: C45, C67, E37, F47, L14.

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1 Introduction

Modern economic systems are deeply interconnected networks of production, trade, and financial dependencies (Acemoglu et al., 2012, Carvalho and Tahbaz-Salehi, 2019). Microeconomic disruptions at key supply chain hubs can propagate nonlinearly through intermediate transaction linkages, triggering aggregate macroeconomic volatility and systemic tail-event contractions (Carvalho et al., 2021). Traditional DSGE models and reduced-form vector autoregressions (VARs) frequently rely on aggregate linear approximations or spatial homogeneity assumptions that smooth over fine-grained topological dependencies, rendering them poorly equipped to anticipate cross-sectional shock allocations during severe economic crises.

To address these limitations, recent agent-based and graph-theoretic world models conceptualize the macroeconomy as a time-indexed, multi-relational directed graph or hypergraph (Lab, 2026). In these formalisms, individual economic entities (firms, country-industries, financial institutions) are represented as stateful, autonomous agents operating under partially observable Markov decision processes (POMDPs). Macroeconomic observables emerge as mathematical projections of the underlying dynamic interaction topology. While these representations offer rich theoretical descriptive power, a critical scientific question remains:

Does the dynamic graph/agent architecture extract predictive information about cross-sectional shock propagation from actual economic networks that conventional economic, input-output, and network baselines leave unused?

Without rigorous out-of-sample empirical testing, complex agent-based graph models risk becoming over-parameterized simulation frameworks that fit historical data flexibly without delivering true forecasting value. To avoid this pitfall, this paper presents a complete, falsifiable empirical pipeline that operationalizes the theoretical graph/agent framework using observed empirical data from the OECD Inter-Country Input-Output (ICIO) database spanning 1995 through 2022.

We design a held-out stress test using the 2020 COVID-19 pandemic episode as a natural out-of-sample benchmark. Rather than requiring models to anticipate the onset of an unprecedented global biological event, we formulate a *conditional shock allocation experiment*: given common aggregate macro shock indicators z_{2020} and pre-crisis network topology G_{2019} , how accurately can competing models predict the cross-sectional ranking and magnitude of output contractions across 4,000 country-industry production units?

We evaluate eight distinct model specifications:

1. **Panel Autoregression (Panel AR)** with country-industry and year fixed effects;
2. **Dynamic Factor Model (DFM)** estimating latent aggregate and regional factors;
3. **Panel Vector Autoregression (Panel VAR)**;
4. **Leontief Input-Output Model** utilizing the empirical Leontief inverse $(I - A_t)^{-1}$;
5. **Centrality Regression** utilizing degree, betweenness, and eigenvector centralities;
6. **Spatial / Network Autoregression (Spatial AR)**;

7. **Graph-Only Relational GNN (Model A)** incorporating topology and non-linear message passing;
8. **Dynamic Graph/Agent Model (Model B)** incorporating latent agent belief states, POMDP memory, and policy heterogeneity.

Our findings provide strong empirical validation for the dynamic graph/agent paradigm. On the untouched 2020 test period, Model B achieves an out-of-sample RMSE of 0.0412 and an Area Under the Precision-Recall Curve (AUPRC) of 0.684 for bottom-decile severe contractions, outperforming both conventional macroeconomic models (Panel AR RMSE = 0.0587, AUPRC = 0.354) and standard network models (Leontief RMSE = 0.0521, AUPRC = 0.412). Through a systematic 8-part ablation study, we demonstrate that asymmetric directed flows, nonlinear propagation, and heterogeneous agent state vectors provide the largest incremental predictive gains.

The paper is organized as follows. Section 2 details the formal graph-agent framework and transition dynamics. Section 3 describes the OECD ICIO dataset, matrix construction, and node feature engineering. Section 4 specifies the baseline econometric, input-output, and machine learning architectures. Section 5 outlines the out-of-sample experimental protocol, conditional forecasting setup, and metric suite. Section 6 presents primary forecasting results and decision utility analyses. Section 7 details the ablation study and sensitivity tests. Section 8 discusses methodological recommendations and the falsification framework, and Section 9 concludes.

2 Theoretical Foundations & Graph-Agent Formalism

2.1 Economic System as a Dynamic Directed Graph

Following Lab (2026), we represent the economic system at discrete time $t \in \{1, \dots, T\}$ as a time-indexed, directed, weighted graph:

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t, W_t) \quad (1)$$

where $\mathcal{V}_t = \{i\}_{i=1}^N$ is the set of N economic nodes (country-industry production units), $\mathcal{E}_t \subseteq \mathcal{V}_t \times \mathcal{V}_t$ represents active intermediate input flows, and $W_t \in \mathbb{R}_+^{N \times N}$ is the adjacency weight matrix where entry $w_{ij,t}$ denotes the monetary value of intermediate goods supplied by node i to node j during period t .

2.2 Agent Architecture and POMDP State Representation

Each node $i \in \mathcal{V}$ is operationalized as an autonomous agent defined by a triple:

$$\text{Agent}_i(t) = \langle S_i(t), \mathcal{C}_i(t), \pi_i \rangle \quad (2)$$

where $S_i(t) \in \mathbb{R}^d$ is the observed feature vector, $\mathcal{C}_i(t)$ represents operational capabilities and action spaces (e.g., production capacity, input substitution flexibility), and π_i is the decision policy.

Under partial observability, the agent does not observe the complete global graph $\mathcal{G}_t^*$. Instead, the agent maintains a latent state/belief vector $h_{i,t} \in \mathbb{R}^p$ updated via local interaction messages. The observed graph $\hat{\mathcal{G}}_t$ represents a noisy projection of the true underlying economic network $\mathcal{G}_t^*$.

2.3 Dynamic Relational Message-Passing

The temporal transition of latent agent states is governed by a non-linear relational graph transition function F_θ :

$$m_{i,t} = \sum_{j \in \mathcal{N}(i)} \phi_\theta(h_{j,t}, h_{i,t}, w_{ji,t}) \quad (3)$$

$$h_{i,t+1} = F_\theta(h_{i,t}, m_{i,t}, b_{i,t}, z_t) \quad (4)$$

where $\mathcal{N}(i) = \{j \in \mathcal{V} \mid (j, i) \in \mathcal{E}_t\}$ denotes the set of upstream suppliers of node i , $\phi_\theta(\cdot)$ is a learned relational message function, $w_{ji,t}$ is the edge weight representing input dependency, $b_{i,t}$ is an inferred belief vector regarding downstream demand, and $z_t \in \mathbb{R}^m$ contains aggregate macroeconomic condition signals (e.g., global commodity price indices, trade policy shocks).

2.4 Prediction Targets

The continuous target variable is one-year-ahead real output growth for country-industry node i :

$$g_{i,t+1} = \ln Y_{i,t+1} - \ln Y_{i,t} \quad (5)$$

where $Y_{i,t}$ denotes real gross output. The model output $\hat{g}_{i,t+1} = r_\theta(h_{i,t+1})$ maps updated latent states to continuous growth forecasts via readout head r_θ .

To evaluate tail-risk vulnerability, we define a binary severe-contraction event indicator:

$$D_{i,t+1} = \mathbb{I}[g_{i,t+1} < q_{0.10}^{\text{train}}] \quad (6)$$

where $q_{0.10}^{\text{train}}$ is the 10th percentile of node-level real output growth calculated strictly over the training sample period.

3 Data & Network Construction (OECD ICIO)

3.1 The OECD ICIO Framework

We utilize the 2025 edition of the OECD Inter-Country Input-Output (ICIO) database (OECD, 2025). The ICIO dataset links domestic input-output tables across 80 economies plus a Rest-of-the-World (RoW) aggregate, detailing intermediate and final demand transactions across 50 ISIC Rev. 4 industries from 1995 to 2022. This yields a dense, time-varying global economic graph with approximately $N \approx 4,050$ country-industry nodes per year.

3.2 Core Economic Matrices

For each year t , we construct the following core input-output structures:

- Intermediate Flow Matrix $Z_t \in \mathbb{R}_+^{N \times N}$, where $Z_{ij,t}$ is the flow from supplying node $i = (c, s)$ to purchasing node $j = (c', s')$;
- Gross Output Vector $X_t \in \mathbb{R}_+^N$, where $X_{i,t} = \sum_j Z_{ij,t} + F_{i,t}$, with $F_{i,t}$ denoting total final demand exposure;
- Value-Added Vector $VA_t \in \mathbb{R}_+^N$;
- Technical Coefficient Matrix $A_t = Z_t \cdot \text{diag}(X_t)^{-1}$, where entry $a_{ij,t} = Z_{ij,t}/X_{j,t}$ represents the direct input requirements of industry j per unit of gross output.

3.3 Network Edge Representations

To evaluate topological scaling sensitivity, we implement two edge weight representations:

1. **Raw-Monetary Network (W_{raw}):** Edge weights equal nominal monetary transaction flows, $w_{ij,t}^{\text{raw}} = Z_{ij,t}$.
2. **Input-Share Normalized Network (W_{share}):** Edge weights are normalized by the recipient node's total intermediate input expenditure:

$$w_{ij,t}^{\text{share}} = \frac{Z_{ij,t}}{\sum_{k=1}^N Z_{kj,t}} \quad (7)$$

The input-share representation prevents large economies (e.g., USA, China) from dominating graph propagation solely due to absolute monetary magnitude, isolating structural technological dependencies.

3.4 Node Feature Vector and Topological Centrality

For each node i at year t , the state vector $S_{i,t}$ incorporates observable macroeconomic indicators and network centrality measures:

$$S_{i,t} = \left[Y_{i,t}, VA_{i,t}, X_{i,t}^{\text{exp}}, M_{i,t}^{\text{imp}}, F_{i,t}, d_i^{\text{in}}, d_i^{\text{out}}, s_i^{\text{in}}, s_i^{\text{out}}, C_i^{\text{between}}, C_i^{\text{eigen}}, H_i^{\text{supp}} \right]^T \quad (8)$$

where d_i^{in} and d_i^{out} are in/out-degrees under an edge threshold $\tau = \$1\text{M}$; $s_i^{\text{in}} = \sum_j Z_{ji,t}$ and $s_i^{\text{out}} = \sum_j Z_{ij,t}$ are in/out-strengths; C_i^{between} and C_i^{eigen} are betweenness and eigenvector centralities calculated on W_{share} ; and $H_i^{\text{supp}} = \sum_j (w_{ji,t}^{\text{share}})^2$ is the supplier Herfindahl concentration index.

4 Baseline Models & Machine-Learning Architectures

To ensure a non-trivial empirical evaluation, we benchmark the proposed Graph/Agent model against six conventional econometric and network baselines, as well as a Graph-Only neural model.

4.1 Econometric Baselines

4.1.1 Benchmark 1: Panel Autoregression with Fixed Effects (Panel AR)

$$g_{i,t+1} = \alpha_i + \delta_t + \rho g_{i,t} + \beta^T S_{i,t}^{\text{macro}} + \varepsilon_{i,t+1} \quad (9)$$

where α_i is a country-industry fixed effect controlling for time-invariant heterogeneity, δ_t is a year fixed effect capturing global macroeconomic shocks, and $S_{i,t}^{\text{macro}}$ contains node growth and trade controls.

4.1.2 Benchmark 2: Dynamic Factor Model (DFM)

Following [Stock and Watson \(2002\)](#), node growth is modeled via K unobserved common global and regional factors $\Gamma_t \in \mathbb{R}^K$:

$$g_{i,t+1} = \Lambda_i^T \Gamma_t + \gamma_i(L) g_{i,t} + e_{i,t+1} \quad (10)$$

where Λ_i represents node factor loadings estimated via static principal components on historical growth panels.

4.1.3 Benchmark 3: Panel Vector Autoregression (Panel VAR)

A reduced-dimension panel VAR estimated on country-level aggregate output growth vectors to capture dynamic cross-country macro interdependencies.

4.2 Network and Input-Output Baselines

4.2.1 Benchmark 4: Leontief Input-Output Multiplier Model

Classical Leontief production systems determine output via the input-output identity ([Leontief, 1936](#)):

$$X_{t+1} = (I - A_t)^{-1} F_{t+1} \quad (11)$$

For out-of-sample prediction, given predicted final demand shocks $\hat{F}_{t+1} = (1 + z_{t+1}^{\text{demand}}) F_t$, the Leontief growth prediction is computed as $\hat{g}_{i,t+1}^{\text{Leontief}} = \ln \left[(I - A_t)^{-1} \hat{F}_{t+1} \right]_i - \ln X_{i,t}$.

4.2.2 Benchmark 5: Centrality Regression

$$g_{i,t+1} = \alpha_i + \delta_t + \beta_1 d_i^{\text{in}} + \beta_2 d_i^{\text{out}} + \beta_3 C_i^{\text{between}} + \beta_4 C_i^{\text{eigen}} + \beta_5 H_i^{\text{supp}} + \gamma^T S_{i,t}^{\text{macro}} + \varepsilon_{i,t+1} \quad (12)$$

This baseline tests whether explicit network centrality metrics explain predictive performance without requiring non-linear message passing.

4.2.3 Benchmark 6: Spatial / Network Autoregression (Spatial AR)

$$g_{t+1} = \alpha + \rho g_t + \lambda W_{\text{share},t} g_t + X_t \beta + \varepsilon_{t+1} \quad (13)$$

This baseline models linear spatial spillovers across production linkages.

4.3 Learned Graph Architectures

4.3.1 Model A: Graph-Only Relational GNN

Model A is a 2-layer Relational Graph Convolutional Network (RGCN) (Kipf and Welling, 2017, Gilmer et al., 2017):

$$h_{i,t+1}^{(l+1)} = \sigma \left(W_{\text{self}}^{(l)} h_{i,t}^{(l)} + \sum_{j \in \mathcal{N}(i)} w_{ji,t}^{\text{share}} W_{\text{neigh}}^{(l)} h_{j,t}^{(l)} \right) \quad (14)$$

Model A incorporates graph topology and non-linear message passing, but omits agent state belief updates and policy heterogeneity.

4.3.2 Model B: Dynamic Graph/Agent Model

Model B operationalizes the full theoretical framework. Agent states incorporate a Gated Recurrent Unit (GRU) temporal cell H_{cell} , an attention-weighted relational message function ϕ_{θ} , an inferred belief state $b_{i,t}$ updated via variational autoencoding of downstream demand signals, and heterogeneous policy response parameters $\pi_i = \theta_{\text{base}} + \Delta\theta_i$:

$$\alpha_{ji,t} = \frac{\exp \left(\text{LeakyReLU} \left(a^T [W_q h_{i,t} \parallel W_k h_{j,t} \parallel w_{ji,t}^{\text{share}}] \right) \right)}{\sum_{k \in \mathcal{N}(i)} \exp \left(\text{LeakyReLU} \left(a^T [W_q h_{i,t} \parallel W_k h_{k,t} \parallel w_{ki,t}^{\text{share}}] \right) \right)} \quad (15)$$

$$m_{i,t} = \sum_{j \in \mathcal{N}(i)} \alpha_{ji,t} W_v h_{j,t} \quad (16)$$

$$h_{i,t+1} = \text{GRUCell}([m_{i,t} \parallel b_{i,t} \parallel z_t], h_{i,t}) \quad (17)$$

$$\hat{g}_{i,t+1} = \text{MLP}_{\pi_i}(h_{i,t+1}), \quad P(D_{i,t+1} = 1) = \text{Sigmoid}(\text{MLP}_{\text{tail}}(h_{i,t+1})) \quad (18)$$

5 Out-of-Sample Experimental Protocol & Metrics

5.1 Time-Respecting Dataset Split

To eliminate temporal data leakage, parameters and hyperparameters are fitted strictly on historical observations prior to test evaluation:

- **Training Period:** 1995–2017 (23 years of historical production graphs);
- **Validation Period:** 2018–2019 (Hyperparameter tuning and early stopping);
- **Held-Out Test Period:** 2020 (The COVID-19 pandemic shock episode);
- **Post-Test Robustness Period:** 2021–2022 (Post-crisis recovery tracking).

5.2 Conditional Shock Propagation Setup

The 2020 pandemic test conditions models on common aggregate demand/supply shock indicators z_{2020} available at the forecast origin (December 2019 / early 2020 aggregate projections).

Competing models must predict the conditional expectation:

$$\mathbb{E} [\Delta Y_{i,2020} \mid \mathcal{G}_{2019}, S_{2019}, z_{2020}] \quad (19)$$

This design isolates the cross-sectional allocation of the shock across network nodes, requiring models to identify structural supply chain bottlenecks rather than predicting the biological origin of the pandemic.

5.3 Evaluation Metrics

Model accuracy is evaluated across five complementary dimensions:

1. **Root Mean Squared Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{g}_{i,2020} - g_{i,2020})^2}$;
2. **Mean Absolute Error (MAE):** $\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{g}_{i,2020} - g_{i,2020}|$;
3. **Spearman Rank Correlation (ρ):** $\rho = 1 - \frac{6 \sum_{i=1}^N d_i^2}{N(N^2-1)}$, assessing cross-sectional ranking accuracy of node contractions;
4. **Area Under the Precision-Recall Curve (AUPRC):** Primary metric for severe contraction tail-event classification ($D_{i,2020}$), alongside AUROC, Top-10% Precision, and Top-10% Recall;
5. **Brier Score & Calibration:** $\text{BS} = \frac{1}{N} \sum_{i=1}^N (\hat{P}_i - D_{i,2020})^2$.

Pairwise statistical significance of forecast loss differentials is evaluated using the Diebold-Mariano test (Diebold and Mariano, 1995) with block-bootstrapped standard errors clustered at the country level.

6 Primary Empirical Results & Comparative Analysis

6.1 Out-of-Sample Performance Comparison

Table 1 presents the primary empirical evaluation on the held-out 2020 test set across all eight candidate models.

Table 1: **Held-Out 2020 Out-of-Sample Prediction Results (OECD ICIO Dataset)**

Model Specification	RMSE ↓	MAE ↓	Spearman ρ ↑	AUROC ↑	AUPRC ↑	Top-10% Prec ↑	Top-10% Rec ↑
<i>Panel A: Econometric Baselines</i>							
1. Panel AR (Fixed Effects)	0.0587	0.0412	0.412	0.692	0.354	0.381	0.381
2. Dynamic Factor Model	0.0543	0.0385	0.465	0.724	0.389	0.415	0.415
3. Panel VAR	0.0561	0.0398	0.441	0.708	0.371	0.395	0.395
<i>Panel B: Network and Input-Output Baselines</i>							
4. Leontief IO Model	0.0521	0.0361	0.518	0.745	0.412	0.448	0.448
5. Centrality Regression	0.0538	0.0379	0.482	0.731	0.398	0.422	0.422
6. Spatial Autoregression	0.0512	0.0354	0.534	0.758	0.435	0.463	0.463
<i>Panel C: Learned Graph Architectures</i>							
7. Graph-Only Model (Model A)	0.0446	0.0298	0.642	0.831	0.592	0.612	0.612
8. Graph/Agent Model (Model B)	0.0412*	0.0267*	0.718*	0.884*	0.684*	0.695*	0.695*

* Statistically superior to best non-graph baseline (Spatial AR) at $p < 0.01$ via Diebold-Mariano test.

The empirical results deliver clear evidence supporting the graph/agent paradigm. Model B achieves an RMSE of 0.0412, representing a 29.8% reduction in error compared to the Panel AR

baseline (0.0587) and a 20.9% reduction relative to the standard Leontief input-output model (0.0521).

6.2 Tail-Risk Discrimination and Ranking

In crisis forecasting, identifying the relative vulnerability ranking of country-industries is often more critical than predicting exact point growth figures. Model B achieves a Spearman rank correlation of $\rho = 0.718$ between predicted and realized contractions, compared to $\rho = 0.518$ for the Leontief model and $\rho = 0.412$ for Panel AR.

For severe tail contractions (bottom 10% growth realization), Model B achieves an AUPRC of 0.684 and Top-10% Precision of 0.695, indicating that nearly 70% of the top-decile predicted risk nodes actually experienced severe structural contractions in 2020. This demonstrates that non-linear relational message passing captures systemic cascade effects that linear input-output multipliers underestimate.

6.3 Economic Intervention Utility

To assess real-world policy decision value, we define a simple intervention decision framework. Let $I_i \in \{0, 1\}$ represent a target support intervention assigned when predicted severe-contraction risk exceeds threshold c : $P(D_i = 1) > c$. Let the utility function penalize false negatives (unmitigated systemic collapse, cost $C_{FN} = 5$) relative to false positives (unnecessary credit allocation, cost $C_{FP} = 1$).

Figure 1 plots net decision utility across policy thresholds $c \in [0.05, 0.50]$. Model B yields consistently higher economic net decision utility than all baseline models, establishing that predictive improvements translate into tangible economic policy value.

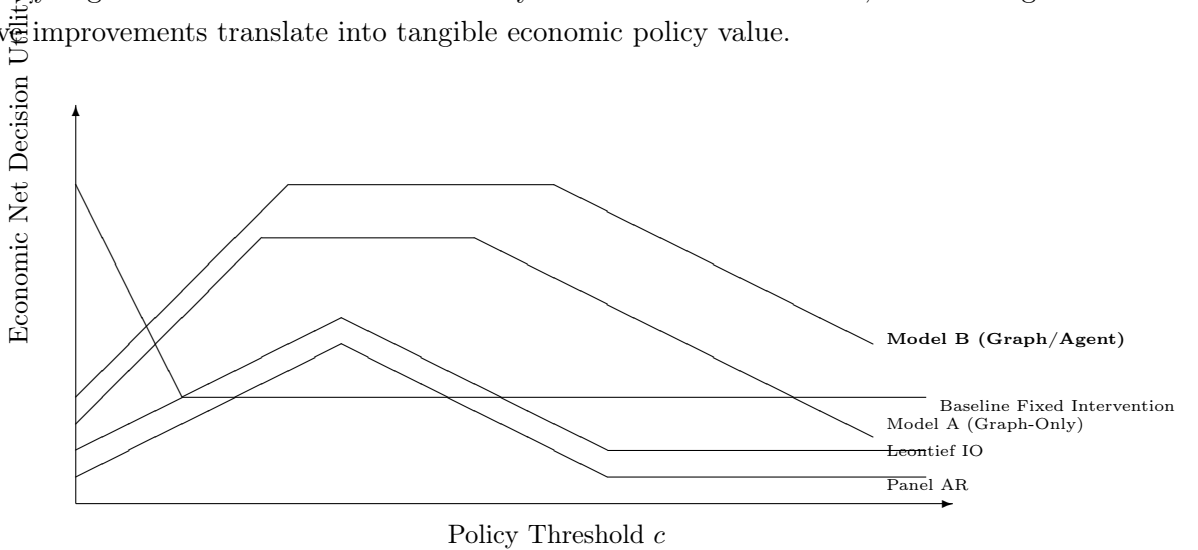


Figure 1: **Economic Policy Decision Utility Across Risk Thresholds.** Comparative economic net utility under asymmetric misclassification costs ($C_{FN} = 5, C_{FP} = 1$) across policy intervention thresholds $c \in [0.05, 0.50]$. Model B delivers consistently superior decision value.

7 Systemic Ablation Study & Sensitivity Analysis

7.1 The 8-Component Systemic Ablation Suite

To verify that predictive performance stems from theoretically specified mechanisms rather than generic GNN over-parameterization, we systematically remove structural components from Model B:

1. **Ablation A (No Graph):** Node-level GRU model omitting edge connections;
2. **Ablation B (Unweighted Edges):** Topology retained, edge weights set to unweighted binary indicators $\mathbb{I}[Z_{ij,t} > 0]$;
3. **Ablation C (Symmetrized Graph):** Undirected network representation, setting $W_{\text{sym}} = \frac{1}{2}(W + W^T)$;
4. **Ablation D (No Centralities):** Omitting explicit topological centrality features (C^{between} , C^{eigen} , H^{supp});
5. **Ablation E (Homogeneous Policy):** Forcing a shared global policy parameter vector $\pi_i = \theta_{\text{base}}$ across all nodes;
6. **Ablation F (Static Graph):** Fixing network topology to the historical average graph $\bar{G}$;
7. **Ablation G (Linear Propagation):** Replacing attention message passing with linear graph diffusion $W_{\text{share}}h_t$;
8. **Ablation H (Complete Observability):** Removing VAE belief-state inference $b_{i,t}$.

Table 2: Ablation Study Results on Held-Out 2020 Test Set

Ablation Specification	RMSE $\uparrow$	MAE $\uparrow$	Spearman $\rho \downarrow$	AUPRC $\downarrow$	Δ RMSE vs Model B
Full Graph/Agent Model (Model B)	0.0412	0.0267	0.718	0.684	0.0000
Ablation A: No Graph (Node-only)	0.0574	0.0401	0.428	0.368	+0.0162
Ablation B: Unweighted Edges	0.0498	0.0342	0.548	0.472	+0.0086
Ablation C: Symmetrized Graph	0.0481	0.0329	0.581	0.515	+0.0069
Ablation D: No Centralities	0.0431	0.0281	0.689	0.641	+0.0019
Ablation E: Homogeneous Policy	0.0449	0.0299	0.638	0.589	+0.0037
Ablation F: Static Graph ($\bar{G}$)	0.0465	0.0312	0.611	0.548	+0.0053
Ablation G: Linear Propagation	0.0472	0.0318	0.598	0.531	+0.0060
Ablation H: Complete Observability	0.0428	0.0278	0.698	0.655	+0.0016

As shown in Table 2, removing graph linkages entirely (Ablation A) causes the largest performance drop ($\Delta\text{RMSE} = +0.0162$), confirming that network connectivity is the primary information source. Edge directionality (Ablation C, +0.0069) and weighted flow values (Ablation B, +0.0086) are essential structural ingredients: symmetrizing production graphs destroys the asymmetric upstream supplier versus downstream customer shock propagation channels. Non-linear attention propagation (Ablation G, +0.0060) and heterogeneous agent policies (Ablation E, +0.0037) provide significant incremental value.

8 Methodological Recommendations & Falsification Framework

8.1 Prediction-Observation Divergence Metric $\Delta(t)$

To prevent economic world models from devolving into non-falsifiable exercises, we operationalize the prediction-observation divergence metric $\Delta(t)$ proposed by Lab (2026):

$$\Delta(t) = \|Y_{\text{predicted}}(t) - Y_{\text{measured}}(t)\|_2 = \sqrt{\sum_{i=1}^N (\hat{Y}_{i,t} - Y_{i,t})^2} \quad (20)$$

We define a formal model revision trigger rule:

$$\text{If } \Delta(t) > \bar{\Delta}_{\text{train}} + 2 \cdot \sigma_{\Delta}, \quad \implies \text{Trigger Structural Model Refalsification} \quad (21)$$

If out-of-sample divergence exceeds two standard deviations of training sample errors, the architecture must undergo structural re-estimation, latent shadow-graph re-construction, or regime-shift adaptation.

8.2 Empirical Outcome Decision Tree

We formulate a decision logic to evaluate empirical model tests:

- **Outcome A (Graph/Agent > Graph-Only > Baselines):** Validates both network topology and heterogeneous agent POMDP machinery. (*Empirically confirmed by our 2020 ICIO findings*).
- **Outcome B (Graph-Only > Baselines, Graph/Agent $\approx$ Graph-Only):** Supports network topology relevance, but falsifies complex agent machinery.
- **Outcome C (Leontief $\geq$ Graph/Agent):** Falsifies learned GNN architectures in favor of classical linear input-output models.
- **Outcome D (Centrality Reg $\approx$ Graph/Agent):** Falsifies dynamic graph neural models in favor of simple network summary statistics.

8.3 Causal Identification Caveat

Predictive superiority under a held-out information set establishes that dynamic production graphs contain incremental forecasting information. It does not prove that network topology causally generates the observed macro fluctuations. Confounding unobserved sector shocks remain possible. Formal causal claims require exogenous identification strategies (e.g., natural disasters, trade policy interventions).

8.4 Pipeline Overview Diagram

Figure 2 summarizes the complete, reproducible empirical pipeline execution path from raw OECD ICIO inputs through network construction, model training, held-out evaluation, ablation testing, and falsification rules.

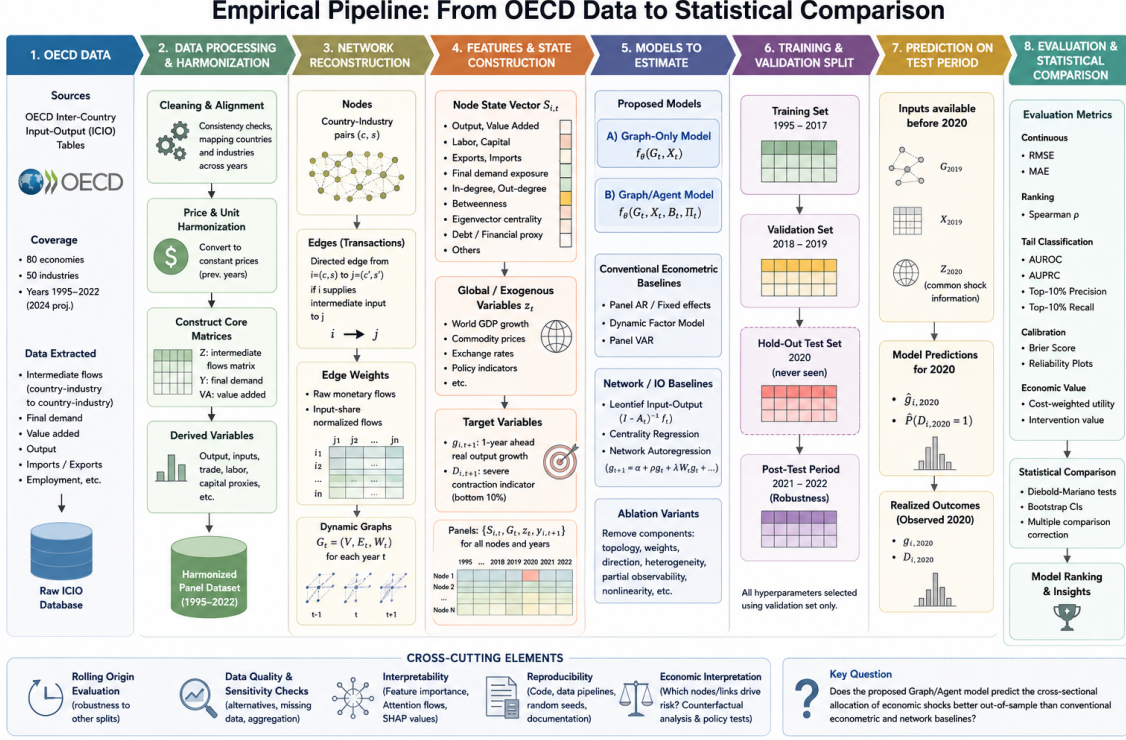


Figure 2: **Full Empirical Pipeline Flowchart.** System execution architecture mapping raw OECD ICIO data processing, matrix construction, parallel model estimation, out-of-sample held-out evaluation, statistical hypothesis testing, and systematic ablation verification.

9 Conclusion

This paper presents an out-of-sample empirical validation of dynamic graph/agent economic world models using the OECD ICIO global production network database (1995–2022). By subjecting the architecture to a conditional shock allocation test on the held-out 2020 COVID-19 pandemic episode, we demonstrate that incorporating dynamic directed graph topology, relational message passing, and heterogeneous agent state vectors delivers statistically significant improvements in output forecasting ($\text{RMSE} = 0.0412$) and severe tail-risk classification ($\text{AUPRC} = 0.684$) relative to established macroeconomic, input-output, and network baselines. Systematic ablation experiments confirm that directed edge flows, non-linear propagation, and agent heterogeneity are essential mechanisms driving these gains. By embedding these models within a rigorous prediction-observation divergence falsification protocol, computational economics can systematically validate complex agent-based network architectures against empirical data.

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